

June 19, 2026

Admissions Committee
Norwegian University of Science and Technology (NTNU)
Ålesund, Norway

Dear Admissions Committee,

I am writing to express my profound interest in the PhD Candidate position focusing on Secure and Trustworthy Data Sharing for Maritime AI Model Development. With a strong background in AI, data engineering, and the mathematical modeling of physical systems, I am highly motivated to tackle the cybersecurity and data-sovereignty bottlenecks preventing the global deployment of Maritime AI.

In my State Engineer thesis ("Gas Turbine Digital Twin"), I engineered predictive maintenance architectures to forecast the degradation of complex industrial machinery. A major challenge in that work was handling highly sensitive telemetry that industrial partners were reluctant to share. To solve similar issues in my Master's thesis ("Cardiovascular Medical Digital Twin"), I developed architectures (Medical_MambaTS and PI-DeepONet) that mathematically enforce physical laws directly into the neural network.

I propose utilizing this background to build a secure, Federated Learning (FL) framework for the Norwegian Maritime AI Center. By training models directly on the edge (aboard ships) and only sharing encrypted parameter updates, we can guarantee data sovereignty. Furthermore, by embedding the physics of maritime vessels (e.g., hydrodynamics) directly into the AI loss function, we can automatically detect and reject cyber-attacks and adversarial data poisoning, as malicious inputs will fundamentally violate physical reality.

I am highly proficient in Python, experienced with complex distributed systems, and eager to collaborate with your industrial partners in Ålesund.

Thank you for your consideration.

Sincerely,
Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>

Research Proposal: Secure and Trustworthy Data Sharing for Maritime AI using Federated Learning and Physics-Informed Digital Twins

1. Introduction and Research Questions

The digitalization of the maritime industry hinges on the deployment of robust Artificial Intelligence (AI) models. However, maritime stakeholders operate in highly competitive and geopolitically sensitive environments. Consequently, the operational data required to train these AI models (e.g., sensor logs, ship design parameters, routing data) is fiercely protected. The central challenge of this PhD project is how to train global, highly accurate maritime AI models without compromising the data sovereignty, cybersecurity, and confidentiality of the contributing partners.

This research proposal addresses the following core research questions: - **RQ1:** How can we construct a decentralized data-sharing architecture that allows maritime stakeholders to collaboratively train AI models without exchanging raw, sensitive operational telemetry? - **RQ2:** How can physics-informed constraints (e.g., fluid dynamics, thermodynamics) be integrated into decentralized AI models to detect adversarial data poisoning and cyber-anomalies? - **RQ3:** What is the quantifiable trade-off between cryptographic security overhead (e.g., homomorphic encryption, differential privacy) and the real-time inference latency required for maritime operations?

2. Theoretical Perspectives

This research draws upon three primary theoretical domains:

2.1. Information Security & Federated Learning (FL)

Rather than aggregating data in a centralized data lake—which creates a massive cybersecurity single point of failure—this project proposes utilizing Federated Learning. In FL, the AI model is distributed to the stakeholders’ edge devices (e.g., on-board ship servers). The model trains locally on the sensitive data, and only the mathematically encrypted weight updates (gradients) are transmitted back to the Norwegian Maritime AI Center. This guarantees data sovereignty, as raw data never leaves the ship.

2.2. Physics-Informed Neural Networks (PINNs) as Anomaly Detectors

A critical vulnerability in decentralized AI is “data poisoning,” where a malicious or compromised node injects fake gradients to corrupt the global model. Drawing from my previous engineering work on Gas Turbine Digital Twins and Medical Digital Twins, I propose using Physics-Informed Neural Networks. Maritime

vessels are governed by strict physical laws (e.g., Navier-Stokes equations for hydrodynamics). By embedding these differential equations into the AI’s loss function, the central server can automatically reject any gradient updates that violate physical reality, effectively nullifying sophisticated cyber-attacks and sensor spoofing.

2.3. Zero-Trust Architecture

The project will be framed within a Zero-Trust security paradigm, assuming that maritime communication channels (e.g., SATCOM) are inherently vulnerable to eavesdropping and manipulation.

3. Methodological Design

The project will be executed through a mixed-methods approach, combining mathematical algorithm design with empirical testing.

Phase 1: Architecture Design and Simulation

- Formulate the mathematical foundation for a secure Federated Learning protocol tailored to maritime bandwidth constraints.
- Implement secure aggregation techniques (e.g., Multi-Party Computation) to protect the gradients from inference attacks.
- Establish the baseline physics-informed loss functions for a maritime test-case (e.g., condition monitoring of propulsion systems).

Phase 2: Cyber-Anomaly Injection and Detection

- Utilize the CPS Lab at NTNU Ålesund to simulate a distributed fleet of vessels.
- Inject deliberate cyber-attacks into the simulation, including False Data Injection (FDI) and adversarial gradient poisoning.
- Evaluate the efficacy of the physics-informed anomaly detection mechanisms in isolating and ignoring compromised nodes.

Phase 3: Industrial Validation

- Deploy the federated architecture in a closed-loop test with industry partners from the Norwegian Maritime AI Center.
- Measure key performance indicators (KPIs) including model convergence speed, communication overhead, and resilience to simulated network partitions.

4. Progress Plan

- **Year 1:** Conduct systematic literature review on maritime cybersecurity and FL. Formalize the mathematical architecture for physics-informed

federated aggregation.

- **Year 2:** Develop the simulation environment. Conduct stress tests against simulated cyber-attacks. Submit findings to a top-tier cybersecurity/AI conference (e.g., IEEE S&P, NeurIPS).
- **Year 3:** Collaborate with maritime user partners (e.g., SINTEF) to deploy a prototype on real-world edge devices. Focus on optimizing the security-latency trade-off. Publish in high-impact journals (e.g., IEEE Transactions on Information Forensics and Security).
- **Year 4:** Finalize the doctoral dissertation, disseminate findings to the Maritime Forum and Blue Maritime clusters, and complete all remaining coursework.

AISSAOUI Ismail

AI & Data Science Engineer

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Portfolio: ismail-aissaoui.vercel.app Location: Chlef, Algeria

LinkedIn: linkedin.com/in/aissaoui-ismail-6a77a92a7 GitHub: github.com/i-aissaoui

PROFILE

AI & Data Science Engineer with a background in Full-Stack Software Engineering. Experienced in building projects across diverse domains, including Industrial Digital Twins, Healthcare, Financial Trading, Computer Vision, and Enterprise Management Systems. My primary interests include Deep Learning, Graph Neural Networks, Reinforcement Learning, and Natural Language Processing (NLP).

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX

AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern)

Jan 2026 – Jul 2026

Sonatrach (Industrial Surveillance Division) — Algeria

Architected an AI-native Digital Twin for real-time monitoring of Gas Turbines.

- Architected a 3-tier Dual-Track framework decoupling safety-critical Edge models from Cloud analytics.
- Developed **TCNEdge** achieving **0.049ms INT8 ONNX C++ inference** for emergency shutdown.
- Engineered **MambaTS** with a missingness-aware gate to maintain robustness during sensor dropouts.
- Implemented **KAN-PCNN** and **ST-KDFormer** to ensure strict thermodynamic consistency at scale.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Data Science

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

ACADEMIC THESES

State Engineering Degree Thesis:

Physics-Informed Explainable Digital Twin for Industrial Gas Turbines: A Dual-Track Cloud-Edge Deep Learning Framework with Koopman-Augmented State-Space Models

Master's Degree Thesis:

Hemodynamic Digital Twins: Fusing Hemodynamic Physics-Informed Neural Networks and Mamba-SSM for ICU Patient Prognostics

PROJECTS

Industrial Gas Turbine Digital Twin (Sonatrach)

2026

- **Edge & Cloud Architecture:** Engineered a 3-tier framework decoupling safety-critical TCNEdge (Siemens PLC) from near-edge MambaTS and cloud-based ST-KDFormer.
- **Ultra-Low Latency Inference:** Compiled 1D Dilated Convolutions (TCNEdge) to INT8 ONNX C++, achieving 0.049ms latency for emergency shutdown.
- **Physcis & Robustness:** Implemented MambaTS with missingness-aware gating for broken sensors, and KAN-PCNN with PyTorch Autograd for absolute thermodynamic consistency.

Tech: MambaTS, TCNEdge, ST-KDFormer, KAN-PCNN, PyTorch, C++ ONNX, PINN

Cardiovascular Medical Digital Twin

2026

- **High-Frequency Edge-Cloud Architecture:** Engineered **Medical_MambaTS** for sub-millisecond bedside inference and dual-branch **SST_KODA_MultiScale** for robust cloud forecasting using 100Hz VitalDB ICU telemetry.
- **Physics-Informed DeepONet:** Implemented a 0D Windkessel **PI_DeepONet** with PyTorch Autograd, forcing the model to obey cardiovascular fluid dynamics without retraining for new patients.
- **Data Pipeline & Robustness:** Developed a PyArrow Parquet zero-copy pipeline avoiding GPU starvation, and designed a custom missingness-aware gate to handle sensor dropouts natively.

Tech: PyTorch, Medical_MambaTS, PI_DeepONet, SST_KODA_MultiScale, Parquet PyArrow, Autograd

Aura — Enterprise AI Trading Platform

2025

- **Hybrid Forecasting:** Built an autonomous trading system combining LSTM networks for time-series predictions.
- **Strategic Decision-Making:** Implemented Deep Q-Learning (Reinforcement Learning) to optimize long-term reward policies.
- **Real-Time Dashboard:** Engineered a high-throughput FastAPI backend with WebSockets and a Next.js interactive UI.

Tech: TensorFlow, PyTorch, FastAPI, Next.js, WebSockets, RL

GNN-based Recruiting System (Career Connect)

2024–2025

- **Relational Learning:** Developed a recommendation engine using Graph Neural Networks (GNN) to map candidate-job relationships.
- **Semantic Embeddings:** Utilized PyTorch Geometric and SBERT for deep semantic text processing.
- **System Architecture:** Deployed via FastAPI and PostgreSQL, significantly outperforming keyword-based heuristics.

Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL, GNN

LANGUAGES

Arabic: Native | English: Advanced | French: Professional

Engineering & Master's Thesis Summaries

Ismail Aissaoui

AI & Data Science Engineer

Master's Thesis: Cardiovascular Medical Digital Twin

Focus: Real-time Hemodynamic Inference, State-Space Models, Physics-Informed Neural Networks

Institution: Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

This thesis proposes the design and implementation of a highly scalable, low-latency **Cardiovascular Medical Digital Twin** capable of performing sub-millisecond inference on high-frequency, multivariate medical telemetry. The core architectural contribution is the **Medical MambaTS** framework, which leverages linear-time state-space models to solve long-sequence dependencies without the quadratic computational bottleneck of traditional Attention mechanisms.

To bridge the gap between purely data-driven black boxes and rigid mathematical models, the thesis introduces the **PI-DeepONet** (Physics-Informed Deep Operator Network). By integrating 0D Windkessel fluid dynamics equations directly into the PyTorch Autograd training loop, the architecture mathematically enforces physical consistency upon the cardiovascular predictions. Furthermore, addressing the clinical reality of disjointed hospital sensors, the system integrates the **SST-KODA-MultiScale** pipeline to robustly impute missing telemetry across non-uniform sampling rates, culminating in a resilient, physics-grounded clinical decision support tool.

State Engineer Thesis: Gas Turbine Digital Twin

Focus: Industrial Predictive Maintenance, Time-Series Forecasting, Thermodynamics

Institution: Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

This thesis focuses on the development of an industrial-grade **Gas Turbine Digital Twin** designed for real-time sensor forecasting and thermodynamic state estimation. The primary objective is to prevent catastrophic failure in heavy machinery by anticipating structural anomalies and sensor drift across highly chaotic, non-stationary thermal environments.

The project benchmarks and deploys a spectrum of advanced sequence modeling architectures, including **LSTMs**, **Transformers**, **Temporal Convolutional Networks (TCN)**, and the hybrid **ST-KDFormer**. The implemented architectures were rigorously evaluated against strict thermodynamic constraints, ensuring that predictions regarding temperature gradients, exhaust velocities, and pressure metrics adhered strictly to the laws of energy conservation. By establishing a direct mapping between raw telemetry signals and impending mechanical degradation, this digital twin serves as a highly reliable predictive maintenance engine for large-scale energy infrastructure.

Academic References

Ismail Aissaoui

AI & Data Science Engineer

Reference 1

Dr. Mohammed Bedjaoui

Primary Thesis Supervisor

Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

Email: m.bedjaoui@esi-sba.dz

Relationship: Supervised both my Master's Thesis (Cardiovascular Medical Digital Twin) and State Engineer Thesis (Gas Turbine Digital Twin). Can comprehensively vouch for my technical research capability, engineering architecture skills, and work ethic.

Reference 2

Dr. Belkacem Khaldi

Academic Professor

Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

Email: b.khaldi@esi-sba.dz

Relationship: Academic professor and course instructor. Can vouch for my theoretical understanding of complex systems and overall academic excellence.

Notice of Pending Graduation

Ismail Aissaoui

Please be advised that I am scheduled to officially defend my Master's and State Engineer theses and graduate on **June 29, 2026**.

My final, official diplomas and certified transcripts will be issued by the Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA) shortly thereafter.

The following pages contain my current academic transcripts to date.